LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	15
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Tracing and identification of various under carriage components, Shock observers, types, brief description and operation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Tracing and identification of various under carriage components, Shock observers, types, brief description and operation
5. Specific Objectives	17.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of shock absorber types on aircraft; Safety: Ensure aircraft is chocked and brakes set; Supervision: Instructor verifies component identification
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the landing gear?
(b) Name two main types of landing gear configurations.
(c) Why is the shock absorber critical during landing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Tracing under carriage components • Identification of shock observers • Types of shock absorbers • Brief description of operation • Construction of alighting gears	L & Ds	PPT, IB	15 min
(b)	• Detailed operation of oleo-pneumatic strut • Comparison of different shock absorber types • Role of wheels and tires in gear • Hydraulic vs mechanical systems • Safety features in landing gear	L & Ds	PPT, PDF	20 min
(c)	• Practical identification of components • Verification of shock absorber types • Review of alighting gear construction • Q&A on operational principles • Clarification of complex mechanisms	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the working principle of an oleo-pneumatic shock absorber.
(b) Differentiate between the main and nose landing gear components.
(c) Describe the sequence of events during gear retraction.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between the shock absorber types discussed and how they relate to the next lesson on wheels.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Wheels, types of wheels and description of typical main wheel	Refer to the master syllabus reference list	LP 16

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min